# Auditing Molecular and Temporal Geometry in AntigenLM Latent Representations of Influenza A HA/NA Sequences

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**Abstract**

Sequence language models are increasingly used as compact representations of rapidly evolving viruses, but their latent geometries should be audited before they are treated as biological or dynamical state spaces. We analyze cached local AntigenLM representations for 111,756 Influenza A HA/NA records from H1N1 and H3N2, each represented as a mean-pooled hidden-state vector $z_i = \mathrm{Enc}_\theta(HA_i, NA_i) \in \mathbb{R}^{384}$. Within subtype, pair-sampled Spearman correlations show that latent Euclidean distance is strongly associated with nucleotide molecular-distance proxies: for HA+NA Hamming distance, mean $\rho$ is 0.8538 in H1N1 and 0.6678 in H3N2 across three random pair-sampling seeds. In contrast, global correlations with absolute temporal separation are weak ($\rho = 0.0612$ and 0.1540), consistent with the fact that influenza evolution is not a one-dimensional chronological trajectory. PCA reveals strong variance concentration (global $n_{95} = 3$, participation ratio 1.91), while TwoNN estimates after exact HA+NA deduplication remain in a low single-digit range across sample sizes and trimming choices. Local latent neighborhoods are temporally enriched relative to subtype-matched random baselines after deduplication: median nearest-neighbor time differences are 2 months versus 42–43 months for H1N1 random pairs and 35 months for H3N2 random pairs. Focused robustness checks show that these molecular correlations persist after exact HA+NA deduplication and that within-subtype permutation of collection month labels removes the short-range temporal enrichment when the latent neighbor graph is held fixed. These results support the narrower claim that the local AntigenLM embedding cloud is molecularly organized under tested sequence-distance proxies, low-dimensional under the tested diagnostics, and locally temporally coherent. They do not establish antigenic similarity, phylogenetic validity, sequence-generation quality, vaccine-strain relevance, or prospective forecasting performance.

## 1 Introduction

Influenza A virus evolves through mutation, reassortment, immune selection, and changing epidemiological context. The surface glycoproteins hemagglutinin (HA) and neuraminidase (NA) are central to surveillance because they shape host-cell entry, viral release, subtype identity, and antigenic drift. However, molecular sequence distance, antigenic phenotype, and future evolutionary success are distinct objects: antigenic cartography and serological assays are needed to measure antigenic distance directly, and sequence similarity alone is not a substitute for those data [20, 1]. Specific HA substitutions can have large antigenic effects [10], which reinforces the need to distinguish molecular proxies from antigenic claims.

Biological sequence language models provide an attractive route for summarizing large collections of viral sequences. Protein language models have shown that unsupervised sequence training can encode structural and functional information [18], and viral protein language models have been used to study evolutionary escape [6]. Genomic language models such as DNABERT, Nucleotide Transformer, and long-context DNA models extend this representation-learning program to nucleotide sequences [7, 3, 16]. AntigenLM is a recent influenza-focused DNA language model designed around functional-unit structure and HA/NA forecasting tasks [17].

Before a learned representation is used as a state space for evolutionary dynamics, its geometry should be checked directly. A useful state-space candidate should at minimum organize molecularly similar records, avoid collapsing into an uninterpretable artifact of time or subtype composition, and exhibit enough low-dimensional structure to make downstream modeling plausible. These criteria are weaker than forecasting validation, but they are necessary safeguards against treating a high-dimensional embedding as biologically meaningful by assertion.

This paper is a geometric audit of cached local AntigenLM embeddings for Influenza A HA/NA records. We ask four descriptive questions. First, does latent Euclidean distance track molecular sequence distance within subtype? Second, does latent distance behave like chronological distance, or is temporal structure primarily local? Third, how concentrated is the embedding variance under PCA? Fourth, do local intrinsic-dimension diagnostics agree qualitatively with the linear PCA picture? We intentionally do not reproduce the AntigenLM forecasting pipeline, generate sequences, optimize mutations, infer vaccine strains, or evaluate prospective forecasts.

# 2 Methods

## 2.1 Data Source and Scope

The analysis uses locally processed Influenza A HA/NA records derived from GISAID data [19]. Records were paired by isolate identifier during preprocessing and retained when HA, NA, subtype, year, and month were available after local quality filters. The current cache contains 111,756 embeddings from records collected between 2000 and 2022: 46,125 H1N1 and 65,631 H3N2 records. Complete sequences are not printed in this manuscript or redistributed in the paper package.

The analysis is limited to the cached HA/NA representations and aggregate metadata. It does not include hemagglutination-inhibition assays, neutralization titers, antigenic map coordinates, Nextstrain clade labels, phylogenetic trees, geographic stratification, host metadata, or prospective post-2022 forecasting outcomes.

## 2.2 Local Embedding Extraction

Each record was encoded with the local AntigenLM-style prediction checkpoint `prediction_sequence/pytorch_model.bin` using a local model/tokenizer wrapper reconstructed from configuration files. The tokenizer represents nucleotides at character level and uses structural tokens for subtype, HA, NA, separator, end-of-sequence, and padding. For a record $i$, the input has the form

$$\langle subtype \rangle \ \langle HA \rangle \ HA_i \ \langle sep \rangle$$
$$\langle NA \rangle \ NA_i \ \langle eos \rangle.$$

The embedding is the attention-mask-weighted mean of the final hidden states,

$$z_i = \mathrm{Enc}_\theta(HA_i, NA_i) \in \mathbb{R}^{384}.$$

The cache metadata records the checkpoint SHA256 hash `6b942a0e2d6af0528a7307ff5754438ad55fdb97390297e9c0f11ffc9803dbff`, maximum sequence length 4000, embedding batch size 4, and no missing cache entries among valid processed records. This local audit should therefore be read as an analysis of the cached representation, not as an official reproduction of all AntigenLM inference or forecasting procedures.

### 2.3 Distances

Latent distance between records $i$ and $j$ is Euclidean:

$$d_Z(i,j) = \|z_i - z_j\|_2.$$

Temporal distance is absolute collection-date separation in months,

$$d_T(i,j) = \left|12(y_i - y_j) + (m_i - m_j)\right|.$$

For a segment sequence $x$ and $y$, normalized Hamming distance is computed over the shared prefix when the length difference is at most 5% of the longer sequence:

$$d_H(x,y) = \frac{1}{L}\sum_{\ell=1}^{L} \mathbf{1}\{x_\ell \neq y_\ell\},$$

where $L$ is the shorter length. Pairs exceeding the length-tolerance rule are omitted for that molecular-distance calculation. HA+NA Hamming distance is a length-weighted average of HA and NA segment distances, with weights given by the shorter HA and NA lengths in the pair. These are molecular sequence-distance proxies. They are not curated biological alignments and are not antigenic distances.

A focused robustness panel also computed NA-only nucleotide Hamming distances and simple translated amino-acid Hamming sensitivity proxies. Amino-acid strings were obtained by frame-0 translation of the available nucleotide strings, with ambiguous codons mapped to an unknown amino-acid symbol and trailing incomplete codons dropped. These translated distances are used only as sensitivity analyses. They should not be interpreted as curated protein alignments, protein-level validation, antigenic-site distances, or antigenic measurements.

### 2.4 Pair-Sampled Correlations

Full all-pairs distance matrices are infeasible at this scale. We therefore sampled 200,000 non-self pairs per subtype for each of three random seeds (42, 7, and 123). All pair sampling was performed within subtype to avoid making subtype separation the dominant source of correlation. For each sampled set, we computed Spearman's rank correlation [21] between $d_Z$ and either $d_T$, HA Hamming distance, or HA+NA Hamming distance. We summarize means and seed-to-seed standard deviations across the three random seeds; these standard deviations are not bootstrap confidence intervals. P-values are not emphasized because pairwise distances are dependent and very large sampled-pair counts can make negligible effects appear statistically significant. The same pair-sampling scale and seeds were reused after exact HA+NA deduplication in the robustness panel.

### 2.5 Deduplication

Exact duplicates are common in viral surveillance data. For TwoNN, local temporal-neighborhood analyses, and robustness pair correlations, we removed exact duplicate HA+NA sequence pairs by retaining the first representative of each exact HA+NA pair encountered in cache order. This left 82,306 records: 36,753 H1N1 and 45,553 H3N2. The primary pair-sampled correlation table uses the full cache, and the robustness table reports deduplicated correlations. This deduplication removes exact HA+NA sequence-pair duplicates only; it does not remove near-duplicates, records duplicated at only one segment, or amino-acid-equivalent nucleotide variants.

### 2.6 PCA Effective Dimension

PCA was computed from the centered embedding matrix using the exact covariance spectrum [8, 9]. We report $n_{80}$, $n_{90}$, $n_{95}$, and $n_{99}$, the number of components required to explain 80%, 90%, 95%, and 99% of variance. We also report the participation ratio,

$$PR = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2},$$

where $\lambda_i$ are positive covariance eigenvalues. PCA effective dimension measures linear variance concentration; it is not identical to local intrinsic dimension or to visualization dimension.

### 2.7 TwoNN Intrinsic-Dimension Diagnostic

TwoNN estimates local intrinsic dimension from ratios of second- to first-nearest-neighbor distances [4]. For each point,

$$\mu_i = \frac{r_{2,i}}{r_{1,i}}, \qquad -\log(1 - F_i) = d \log(\mu_i),$$

where $r_{1,i}$ and $r_{2,i}$ are nearest-neighbor distances after standardizing embedding coordinates, and $F_i$ is the empirical cumulative distribution of $\mu$. We evaluated sample sizes of 5,000, 10,000, 20,000, and 50,000 after exact HA+NA deduplication, with seeds 42, 7, and 123 and trimming levels 0.01 and 0.05. The result is interpreted as a sensitivity range, not as an exact manifold dimension. Alternative estimators such as Levina-Bickel maximum-likelihood methods and DANCo are discussed as future robustness checks [12, 2].

### 2.8 Local Temporal Neighborhoods

For each subtype after exact HA+NA deduplication, we computed Euclidean nearest neighbors in latent space for $k = 5, 10, 20$. Self-matches were excluded by requesting $k + 1$ neighbors and removing the first returned neighbor. We compared absolute month-index differences between each record and its top-$k$ latent neighbors against subtype-matched random pairs with the same number of comparisons. As a negative-control variant, we held the latent neighbor graph fixed but permuted collection month-index labels within subtype across three seeds. The random and date-permutation baselines control for subtype-specific date distributions but not for all possible sources of sampling bias, such as geography, host, collection intensity, lineage composition, or sequence length.

## 3 Results

### 3.1 Dataset Composition

The cached dataset contains 111,756 embeddings of dimension 384, with 46,125 H1N1 and 65,631 H3N2 records. The processed records are unevenly distributed across years, reflecting surveillance intensity and subtype-specific sampling histories (Figure 1). HA and NA length distributions are concentrated near expected segment lengths but include subtype-specific and record-level variation (Figure 2). Exact HA+NA duplicates are frequent: 9,372 H1N1 records and 20,078 H3N2 records are removed by exact HA+NA deduplication.

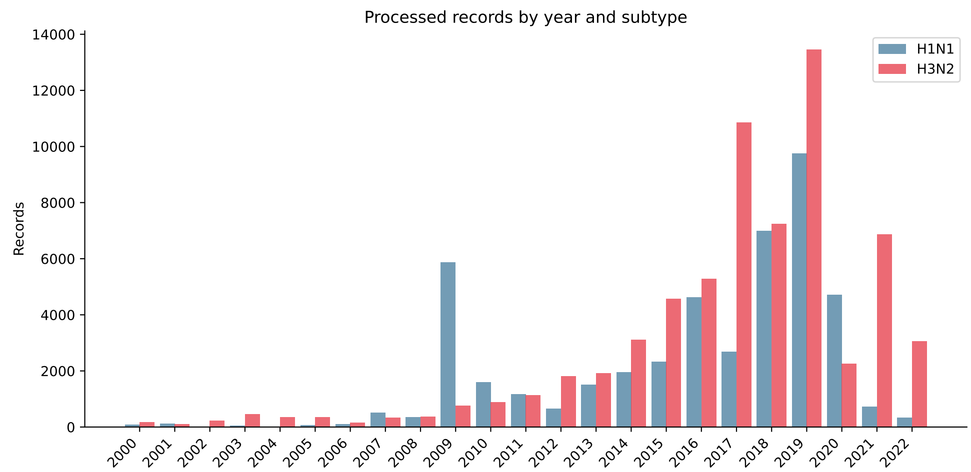



Figure 1: Processed records by year and subtype. Counts reflect the local processed GISAID-derived HA/NA dataset used to build the cache.

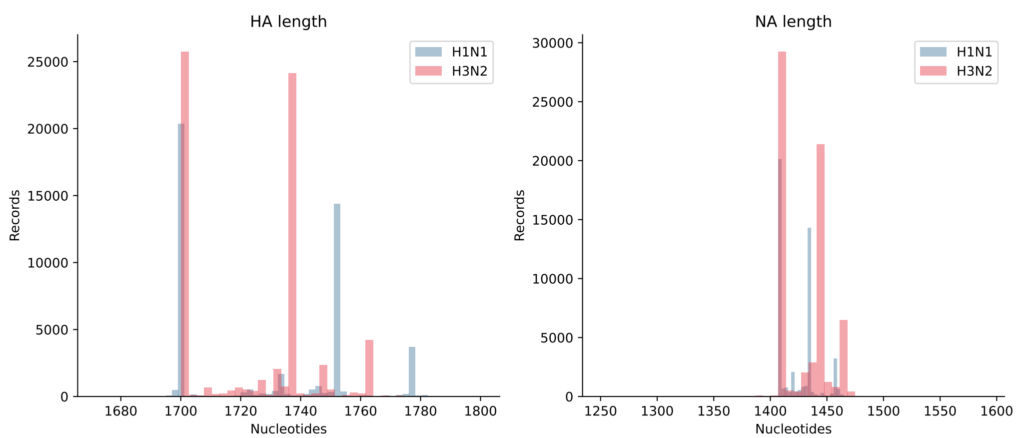



Figure 2: HA and NA nucleotide sequence-length distributions by subtype after local preprocessing and quality filtering.

### 3.2 Latent Distance Tracks Molecular Distance Much More Strongly Than Global Time

Within subtype, latent Euclidean distance is strongly associated with nucleotide Hamming distance, especially for the combined HA+NA proxy (Table 1, Figure 3). HA+NA Hamming correlations are 0.8538 for H1N1 and 0.6678 for H3N2. HA-only correlations are also substantial: 0.8280 for H1N1 and 0.6082 for H3N2. These values support a molecular organization claim for the local embedding geometry.

By contrast, correlations with absolute temporal separation are weak: 0.0612 for H1N1 and 0.1540 for H3N2. This does not imply absence of evolutionary structure. A branching, reassorting, spatially sampled viral population should not be expected to lie on a single chronological line in latent Euclidean distance. The more relevant question for downstream local modeling is whether neighborhoods, rather than all pairs, are temporally coherent.

Table 1: Pair-sampled Spearman correlations between latent Euclidean distance and temporal or molecular sequence-distance proxies. Values summarize three random pair-sampling seeds, with 200,000 requested pairs per subtype per seed; reported standard deviations are seed-to-seed standard deviations, not confidence intervals. Molecular pairs may be omitted when the length-tolerance rule fails.

| Metric | Subtype | $\rho$ mean | $\rho$ sd | valid pairs |
|---|---|---|---|---|
| Temporal distance | H1N1 | 0.0612 | 0.0009 | 200000 |
| Temporal distance | H3N2 | 0.1540 | 0.0008 | 200000 |
| HA Hamming | H1N1 | 0.8280 | 0.0009 | 199914 |
| HA Hamming | H3N2 | 0.6082 | 0.0005 | 199989 |
| HA+NA Hamming | H1N1 | 0.8538 | 0.0006 | 199903 |
| HA+NA Hamming | H3N2 | 0.6678 | 0.0014 | 199538 |

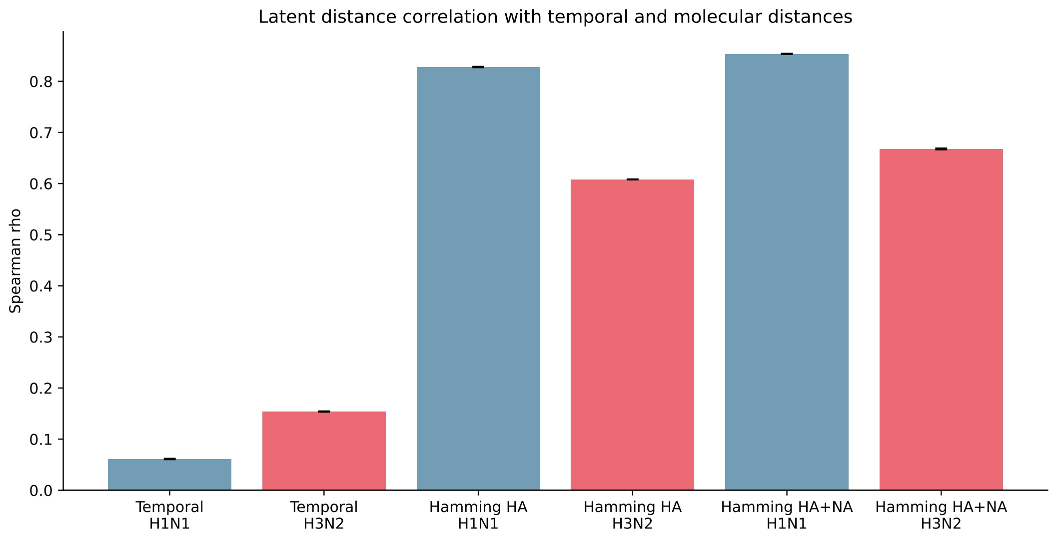



Figure 3: Spearman correlation summary for latent Euclidean distance versus temporal and molecular sequence-distance proxies. Correlations are computed within subtype and summarized across random pair-sampling seeds.

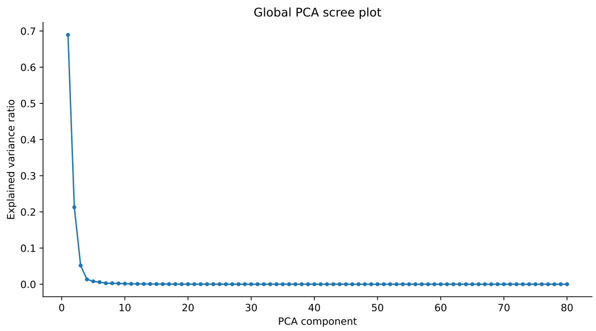
### 3.3 PCA Shows Strong Linear Variance Concentration

The global embedding spectrum is highly concentrated (Table 2, Figure 4). Two components explain at least 90% of global variance, three explain at least 95%, and 10 explain at least 99%. The global participation ratio is 1.91. Subtype-specific spectra are also concentrated, with H1N1 reaching 95% variance in three components and H3N2 in five components.

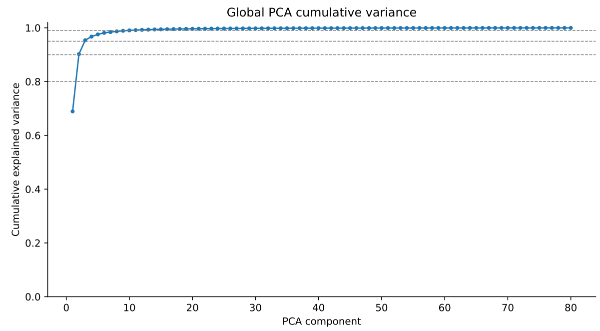
This result should be interpreted as strong anisotropy and low linear effective dimension. It does not, by itself, prove that the data lie on a smooth low-dimensional biological manifold. The first PCs may reflect a mixture of molecular, subtype-specific, temporal, sampling-density, and model-induced directions.

Table 2: PCA effective dimension of AntigenLM embeddings. $n_q$ denotes the number of principal components required to explain fraction $q$ of variance; PR is the participation ratio of the covariance spectrum.

| Group | n | $n_{80}$ | $n_{90}$ | $n_{95}$ | $n_{99}$ | PR |
|---|---|---|---|---|---|---|
| global | 111,756 | 2 | 2 | 3 | 10 | 1.91 |
| H1N1 | 46,125 | 1 | 2 | 3 | 11 | 1.43 |
| H3N2 | 65,631 | 1 | 3 | 5 | 16 | 1.53 |

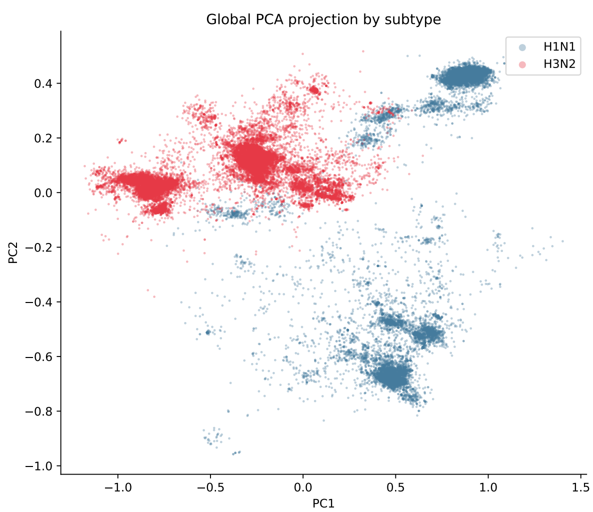



(a) Global scree plot.

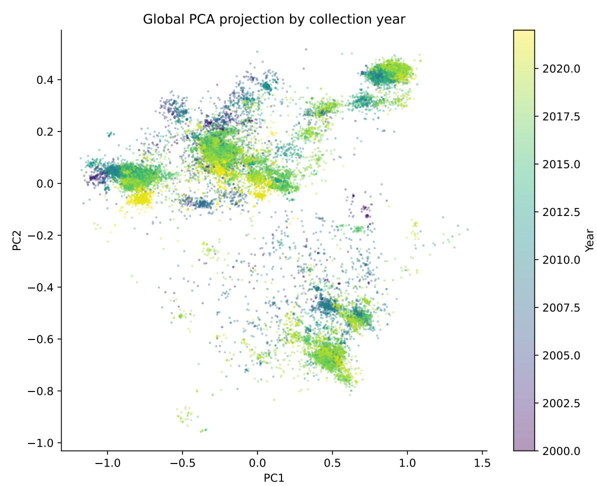



(b) Global cumulative variance.

Figure 4: Global PCA spectrum. The spectrum is sharply concentrated, supporting low linear effective dimension but not establishing intrinsic manifold dimension.




(a) Colored by subtype.




(b) Colored by year.

Figure 5: Two-dimensional global PCA projection. The projection is descriptive and should not be read as a complete representation of evolutionary or antigenic relationships.

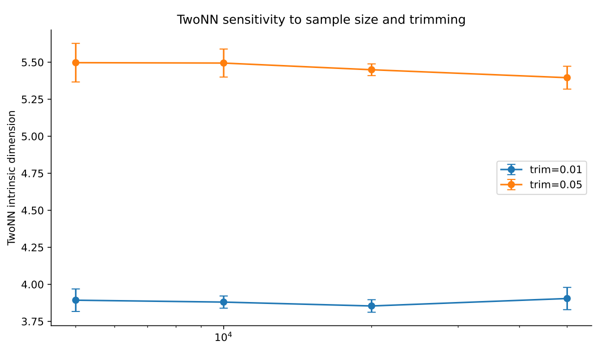
### 3.4 TwoNN Estimates Are Low but Sensitive to Trimming

TwoNN provides a local nearest-neighbor diagnostic that is complementary to the linear PCA spectrum. After exact HA+NA deduplication, estimated dimension remains stable across sample sizes but changes with trimming level (Table 3, Figure 6). With trim 0.01, estimates are approximately 3.85–3.90; with trim 0.05, they are approximately 5.40–5.50. Mean $R^2$ values are near 0.97–0.98.

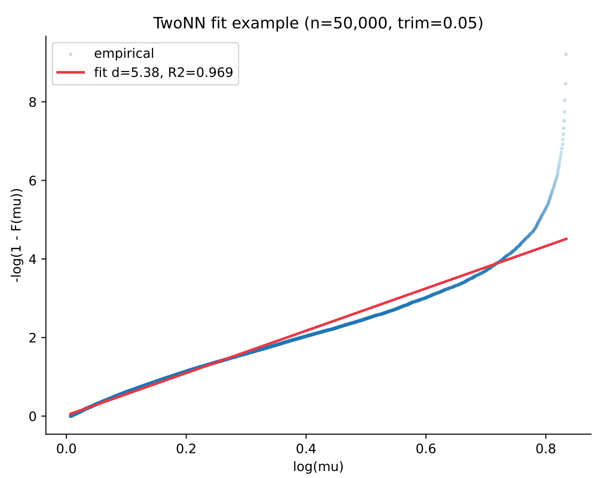
The key conclusion is not a single exact dimension. Rather, PCA and TwoNN agree qualitatively that the nominal 384-dimensional representation has much lower effective structure under the tested preprocessing choices. The trim sensitivity is informative: intrinsic-dimension estimates should be reported as diagnostics with assumptions, not as fixed biological constants.

Table 3: TwoNN intrinsic-dimension sensitivity after exact HA+NA deduplication. Values summarize three random seeds for each sample size and trim level.

| Sample size | trim | dimension mean | $R^2$ mean |
|---|---|---|---|
| 5,000 | 0.01 | 3.89 | 0.981 |
| 5,000 | 0.05 | 5.50 | 0.972 |
| 10,000 | 0.01 | 3.88 | 0.978 |
| 10,000 | 0.05 | 5.49 | 0.972 |
| 20,000 | 0.01 | 3.85 | 0.978 |
| 20,000 | 0.05 | 5.45 | 0.971 |
| 50,000 | 0.01 | 3.90 | 0.980 |
| 50,000 | 0.05 | 5.40 | 0.970 |




(a) Sensitivity to sample size and trimming.




(b) Example linear fit.

Figure 6: TwoNN intrinsic-dimension diagnostics after exact HA+NA deduplication. Estimates should be interpreted as a sensitivity range because nearest-neighbor intrinsic-dimension methods are affected by density, anisotropy, trimming, and finite-sample effects.

### 3.5 Temporal Structure Is Local Rather Than Globally Chronological

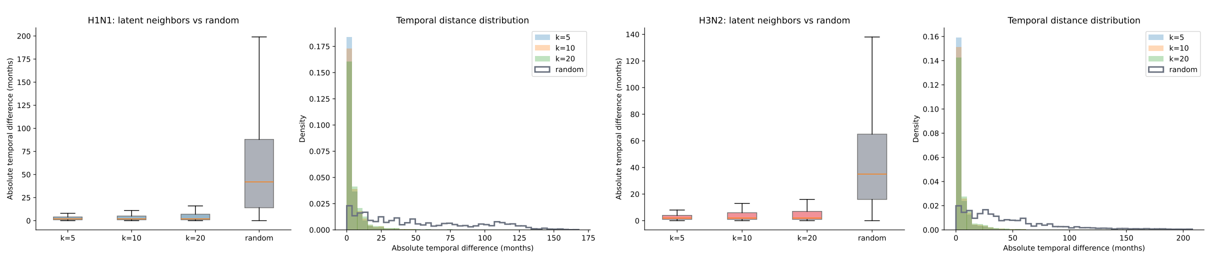
Although global temporal correlations are weak, latent nearest neighbors are much closer in time than subtype-matched random pairs after exact HA+NA deduplication (Table 4, Figure 7). For

H1N1, median time difference among latent neighbors is 2 months across $k = 5, 10, 20$, compared with 42–43 months for random subtype-matched pairs. For H3N2, the corresponding medians are 2 months for latent neighbors and 35 months for random pairs.

This result supports local temporal coherence in the embedding cloud. It does not show that latent space is a forecasting model, nor does it rule out sampling-density effects. It does indicate that the molecular organization seen in the distance correlations is accompanied by short-range temporal enrichment in local neighborhoods.

Table 4: Local temporal coherence of latent nearest neighbors after exact HA+NA deduplication. Random baselines are subtype-matched and use the same number of comparisons as the corresponding neighbor set.

| Subtype | k | median NN months | median random months | ratio |
|---|---|---|---|---|
| H1N1 | 5 | 2.00 | 43.00 | 0.047 |
| H1N1 | 10 | 2.00 | 42.00 | 0.048 |
| H1N1 | 20 | 2.00 | 42.00 | 0.048 |
| H3N2 | 5 | 2.00 | 35.00 | 0.057 |
| H3N2 | 10 | 2.00 | 35.00 | 0.057 |
| H3N2 | 20 | 2.00 | 35.00 | 0.057 |




(a) H1N1. (b) H3N2.

Figure 7: Temporal differences of latent nearest neighbors versus subtype-matched random baselines after exact HA+NA deduplication.

### 3.6 Focused Robustness Checks Support the Main Geometry Claims

Exact HA+NA deduplication reduces but does not remove the molecular-distance signal (Table 5). After deduplication, HA+NA nucleotide Hamming correlations remain high: 0.8358 for H1N1 and 0.6315 for H3N2. NA-only nucleotide correlations are lower than HA-only correlations in both subtypes, while the combined HA+NA nucleotide proxy remains the strongest nucleotide-distance summary. The simple translated amino-acid HA+NA sensitivity proxy also remains substantially correlated with latent distance, although it should be interpreted cautiously because it is not based on curated protein alignments.

Table 5: Focused robustness panel after exact HA+NA deduplication. Values are pair-sampled Spearman correlations summarized across seeds 42, 7, and 123 with 200,000 requested pairs per subtype per seed; reported standard deviations are seed-to-seed standard deviations, not confidence intervals. Amino-acid distances use simple frame-0 translation and are included only as sensitivity proxies, not curated protein alignments.

| Metric | Subtype | $\rho$ mean | $\rho$ sd |
|---|---|---|---|
| Temporal distance | H1N1 | 0.0690 | 0.0031 |
| Temporal distance | H3N2 | 0.1497 | 0.0017 |
| Nucleotide HA Hamming | H1N1 | 0.8086 | 0.0013 |
| Nucleotide HA Hamming | H3N2 | 0.5883 | 0.0006 |
| Nucleotide NA Hamming | H1N1 | 0.7677 | 0.0010 |
| Nucleotide NA Hamming | H3N2 | 0.5287 | 0.0025 |
| Nucleotide HA+NA Hamming | H1N1 | 0.8358 | 0.0010 |
| Nucleotide HA+NA Hamming | H3N2 | 0.6315 | 0.0015 |
| Translated amino-acid HA+NA proxy | H1N1 | 0.7563 | 0.0013 |
| Translated amino-acid HA+NA proxy | H3N2 | 0.6131 | 0.0008 |

The temporal-neighborhood result is also robust to a date-label permutation control (Table 6). Holding the latent neighbor graph fixed but permuting collection month-index labels within subtype increases median neighbor time differences from 2 months to 42–43 months in H1N1 and 35 months in H3N2, matching the subtype-random baseline. This supports the interpretation that local temporal coherence is tied to the observed collection dates rather than to the marginal date distribution alone, under this specific within-subtype permutation control.

Table 6: Date-label permutation control for local temporal neighborhoods after exact HA+NA deduplication. The latent neighbor graph is fixed and collection month-index labels are permuted within subtype across seeds 42, 7, and 123.

| Subtype | k | true median months | permuted median months | true/permuted |
|---|---|---|---|---|
| H1N1 | 5 | 2.00 | 42.33 | 0.047 |
| H1N1 | 10 | 2.00 | 42.67 | 0.047 |
| H1N1 | 20 | 2.00 | 42.67 | 0.047 |
| H3N2 | 5 | 2.00 | 35.00 | 0.057 |
| H3N2 | 10 | 2.00 | 35.00 | 0.057 |
| H3N2 | 20 | 2.00 | 35.00 | 0.057 |

# 4 Discussion

This audit supports a cautious but useful interpretation of the local AntigenLM embedding cloud. The strongest result is molecular: latent Euclidean distance is highly rank-associated with HA and HA+NA nucleotide Hamming distances within subtype. This indicates that the representation is not merely a high-dimensional black box for these records; its metric geometry captures a substantial amount of input-sequence variation.

The robustness panel strengthens this interpretation within the limits of the controls performed. Molecular correlations persist after exact HA+NA deduplication, and the pattern is not restricted to HA alone: NA-only correlations are also substantial, although lower than HA-only in this panel. The simple translated amino-acid sensitivity proxy gives a qualitatively similar result, while remaining less definitive than a curated protein-alignment analysis.

The temporal results clarify what kind of evolutionary structure is present. Global chronological distance is weakly correlated with latent distance, which is expected for a branching and unevenly sampled viral population. Local neighborhoods, however, are temporally coherent after exact HA+NA deduplication. The within-subtype date-label permutation control moves median neighbor time differences back to random-like values when the latent neighbor graph is held fixed, indicating that the local temporal signal depends on the observed temporal labels rather than only on the marginal collection-date distribution. For future dynamical modeling, this local result is more relevant than global correlation with time, because a state-space model would usually rely on local transitions or local neighborhoods rather than treating all pairs of years as lying along one line.

PCA and TwoNN give complementary evidence of low effective structure. PCA shows that variance is concentrated in a few dominant directions, while TwoNN suggests low local dimension under standardized nearest-neighbor ratios. These are encouraging diagnostics for reduced modeling, but they are not a mathematical proof of a smooth manifold or of the correct dimension for a downstream dynamical model.

The results are best described as validating a narrow geometric precondition: the cached embeddings are molecularly organized, anisotropic/low-dimensional under the tested diagnostics, and locally temporally enriched. That is enough to motivate further work, but not enough to claim antigenic prediction, vaccine relevance, or prospective forecasting.

# 5 Limitations

The most important limitation is biological. Nucleotide Hamming distance is a molecular sequence-similarity proxy, not a curated alignment, protein structural comparison, or experimental measure of antigenic distance. This paper does not include curated multiple sequence alignments, HI assay titers, neutralization measurements, antigenic cartography coordinates, clade labels, phylogenetic distances, or fitness measurements. Therefore, it cannot establish antigenic similarity, immune escape, clade success, phylogenetic validity, or vaccine-strain relevance.

The second limitation is methodological. Spearman correlations summarize global rank association over sampled pairs, but pairwise distances are dependent and do not fully assess local neighborhood fidelity. The reported standard deviations are seed-to-seed summaries across three random seeds, not bootstrap confidence intervals. The deduplicated and date-permutation controls reduce two obvious concerns, but no random-embedding, matched-spectrum random-embedding, PCA-whitened, or untrained-checkpoint control is included. Additional diagnostics such as Mantel-style tests, distance correlation, kNN retrieval precision, local distance-quantile analyses, and trustworthiness/continuity would provide a stronger metric validation framework [14, 22, 11]. These broader rank-preservation analyses are not currently present in the repository results.

The translated amino-acid robustness analysis is also limited. It uses simple frame-0 translation of the available nucleotide strings, maps ambiguous codons to an unknown symbol, and drops trailing incomplete codons. It is a sensitivity analysis for whether the broad molecular signal persists under a crude translated representation; it is not curated protein extraction, multiple sequence alignment, antigenic-site analysis, or protein-level validation.

The third limitation concerns intrinsic dimension. PCA effective dimension can be driven by anisotropy, subtype structure, sampling imbalance, or model-induced dominant directions. TwoNN can be affected by exact and near duplicates, nonuniform density, boundary effects, finite samples, coordinate standardization, and trimming. The current results should therefore be read as evidence for low-dimensional structure under specific diagnostics, not as an exact intrinsic dimension estimate.

The fourth limitation is surveillance bias. GISAID-derived sequence availability varies by year, location, laboratory practice, host, and public-health context. The current subtype-matched random baseline does not control for geography, host, collection intensity, or lineage composition. Local temporal coherence may partly reflect bursts of dense sampling around outbreaks or surveillance campaigns.

Finally, this is not a reproduction of the full AntigenLM forecasting protocol. It does not decode or generate sequences, score candidate mutations, compare against WHO vaccine recommendations, evaluate post-2022 prospective forecasts, or benchmark against phylogenetic forecasting methods such as fitness models or tree-shape predictors [13, 15, 5].

# 6 Reproducibility and Data Availability

The local analysis pipeline is script-based. Embeddings were produced by `latent_geometry.py`, full-data metrics and figures by `latent_geometry_full_analysis.py`, and the paper package by `make_paper_1_latent_geometry_full.py`. The focused robustness panel was run with `paper_revision_outputs/run_robustness_panel.py`. The primary aggregate results are stored in `results/latent_geometry_full_metrics.json` and summarized in `results/latent_geometry_full_summary.md`; robustness results are stored in `paper_revision_outputs/robustness_panel_results.json` and `paper_revision_outputs/robustness_panel_summary.md`. The manuscript tables and figures are generated from aggregate outputs and copied into the paper directory.

Raw sequence redistribution may be restricted by the terms of the original data source. For that reason, this manuscript reports aggregate metrics, counts, tables, and figures rather than complete sequences. A submission version should provide public code and aggregate outputs where permitted, plus instructions for authorized data users to rebuild the processed dataset and embedding cache. The local AntigenLM checkpoint provenance, access path, and checksum should be documented explicitly for submission. Any final data-availability statement should also follow the applicable GISAID acknowledgement and data-use terms, without redistributing restricted full sequences.

# 7 Ethics, Biosafety, and Responsible Use

This work is an analysis of existing influenza sequence embeddings. It does not generate viral sequences, optimize mutations, synthesize constructs, provide experimental protocols, make vaccine recommendations, or advise clinical or public-health decisions. Any future use of learned representations for forecasting, strain prioritization, or antigenic interpretation should be evaluated with appropriate prospective validation, domain review, and biosafety governance.

# 8 Conclusion

The cached local AntigenLM representations of Influenza A HA/NA records show strong molecular organization under nucleotide Hamming proxies, weak global chronological association, clear local temporal-neighborhood enrichment, and low effective dimension under PCA and TwoNN diagnostics. Focused robustness checks show that molecular correlations persist after exact HA+NA deduplication and that within-subtype date-label permutation removes the observed local temporal enrichment when the latent neighbor graph is held fixed. These findings make the embedding cloud a plausible candidate state space for future probabilistic modeling. They should be treated as a geometric precondition, not as evidence of antigenic validity, sequence-generation reliability, vaccine-strain relevance, or prospective forecasting skill.

## References


[1] Trevor Bedford, Marc A. Suchard, Philippe Lemey, Gytis Dudas, Victoria Gregory, Alan J. Hay, John W. McCauley, Colin A. Russell, Derek J. Smith, and Andrew Rambaut. Integrating influenza antigenic dynamics with molecular evolution. *eLife*, 3:e01914, 2014.

[2] Claudio Ceruti, Simone Bassis, Alessandro Rozza, Gabriele Lombardi, Elena Casiraghi, and Paola Campadelli. DANCo: An intrinsic dimensionality estimator exploiting angle and norm concentration. *Pattern Recognition*, 47(8):2569–2581, 2014.

[3] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, Hassan Sirelkhatim, Guillaume Richard, Marcin Skwark, Karim Beguir, Marie Lopez, and Thomas Pierrot. Nucleotide transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 22:287–297, 2025.

[4] Elena Facco, Maria d'Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7:12140, 2017.

[5] James Hadfield, Colin Megill, Sidney M. Bell, John Huddleston, Barney Potter, Charlton Callender, Pavel Sagulenko, Trevor Bedford, and Richard A. Neher. Nextstrain: Real-time tracking of pathogen evolution. *Bioinformatics*, 34(23):4121–4123, 2018.

[6] Brian Hie, Ellen D. Zhong, Bonnie Berger, and Bryan Bryson. Learning the language of viral evolution and escape. *Science*, 371(6526):284–288, 2021.

[7] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V. Davuluri. DNABERT: Pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021.

[8] I. T. Jolliffe. *Principal Component Analysis*. Springer, 2 edition, 2002.

[9] Ian T. Jolliffe and Jorge Cadima. Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A*, 374(2065):20150202, 2016.

[10] Bjorn F. Koel, David F. Burke, Theo M. Bestebroer, Stefan van der Vliet, Gerben Zondag, Gaby Vervaet, Eugene Skepner, Nicola S. Lewis, Monique I. Spronken, Colin A. Russell, Mikhail Y. Eropkin, Aeron C. Hurt, Ian G. Barr, Jan C. de Jong, Guus F. Rimmelzwaan, Albert D. M. E. Osterhaus, Ron A. M. Fouchier, and Derek J. Smith. Substitutions near the receptor binding site determine major antigenic change during influenza virus evolution. *Science*, 342(6161):976–979, 2013.

[11] John A. Lee and Michel Verleysen. Quality assessment of dimensionality reduction: Rank-based criteria. *Neurocomputing*, 72(7–9):1431–1443, 2009.

[12] Elizaveta Levina and Peter J. Bickel. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems 17*, pages 777–784, 2005.

[13] Marta Łuksza and Michael Lässig. A predictive fitness model for influenza. *Nature*, 507:57–61, 2014.

[14] Nathan Mantel. The detection of disease clustering and a generalized regression approach. *Cancer Research*, 27(2):209–220, 1967.

[15] Richard A. Neher, Colin A. Russell, and Boris I. Shraiman. Predicting evolution from the shape of genealogical trees. *eLife*, 3:e03568, 2014.

[16] Eric Nguyen, Michael Poli, Matthew G. Durrant, Armin W. Thomas, Brian Kang, Jeremy Sullivan, Madelena Y. Ng, Ashley Lewis, Aman Patel, Aaron Lou, Stefano Ermon, Stephen A. Baccus, Tina Hernandez-Boussard, and Christopher Ré. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, 2023.

[17] Yue Pei, Xuebin Chi, and Yu Kang. AntigenLM: Structure-aware DNA language modeling for influenza. In *International Conference on Learning Representations*, 2026. Accepted by ICLR 2026.

[18] Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15):e2016239118, 2021.

[19] Yuelong Shu and John McCauley. GISAID: Global initiative on sharing all influenza data – from vision to reality. *Euro Surveillance*, 22(13):30494, 2017.

[20] Derek J. Smith, Alan S. Lapedes, Jan C. de Jong, Theo M. Bestebroer, Guus F. Rimmelzwaan, Albert D. M. E. Osterhaus, and Ron A. M. Fouchier. Mapping the antigenic and genetic evolution of influenza virus. *Science*, 305(5682):371–376, 2004.

[21] Charles Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.

[22] Gábor J. Székely, Maria L. Rizzo, and Nail K. Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, 35(6):2769–2794, 2007.